THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The point $(5, -2)$ is reflected in the y-axis. The image is: 1

(a) $(5, -2)$ **(b)** $(-2, -5)$ **(c)** $(-5, -2)$ **(d)** $(-5, 2)$

2. Which of these is a linear polynomial? 1

(a) 7 **(b)** $x^3 - x$ **(c)** $3x + 4$ **(d)** $x^2 + 4$

3. The number of decimal places in the expansion of $\frac{3}{2^3 \times 5^2}$ is: 1

(a) 5 **(b)** 2 **(c)** 4 **(d)** 3

4. $(2x + 3y)^2 =$ 1

(a) $4x^2 + 6xy + 9y^2$ **(b)** $4x^2 + 12xy + 9y^2$ **(c)** $4x^2 - 12xy + 9y^2$ **(d)** $4x^2 + 9y^2$

5. The distance between $(1, 2)$ and $(4, 6)$ is: 1

(a) 5 **(b)** 7 **(c)** 4 **(d)** 6

6. If $x = 5$ is a zero of the polynomial $2x + k$, then $k =$ 1

(a) -11 **(b)** -9 **(c)** -10 **(d)** -8

7. $x = 0.\overline{27}$ equals: 1

(a) $\frac{4}{11}$ **(b)** $\frac{3}{11}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{11}$

8. $(3x - 2y)^2 =$ 1

(a) $9x^2 - 12xy + 4y^2$ **(b)** $9x^2 + 12xy + 4y^2$ **(c)** $9x^2 + 4y^2$ **(d)** $9x^2 - 6xy + 4y^2$

9. The midpoint of the segment joining $(-4, 6)$ and $(8, -2)$ is: 1

(a) $(2, 2)$ **(b)** $(-2, -2)$ **(c)** $(-2, 2)$ **(d)** $(2, -2)$

10. The value of $3x + 5$ at $x = -2$ is: 1

(a) -2 **(b)** 0 **(c)** 1 **(d)** -1

11. An irrational number between 2 and 3 is: 1

(a) $\sqrt{3}$ **(b)** $\sqrt{5}$ **(c)** $\sqrt{10}$ **(d)** $\sqrt{11}$

12. $(x + 7)(x - 3) =$ 1

(a) $x^2 - 4x - 21$ **(b)** $x^2 + 4x - 21$ **(c)** $x^2 + 10x - 21$ **(d)** $x^2 + 4x + 21$

13. A point on the x-axis at distance 6 units to the right of the origin has coordinates: 1

(a) $(7, 0)$ **(b)** $(6, 0)$ **(c)** $(-6, 0)$ **(d)** $(0, 6)$

14. The line $y = 3x + 1$ passes through $(2, p)$. The value of p is: 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 9

15. The sum of a rational number and an irrational number is: 1

(a) zero **(b)** an integer **(c)** rational **(d)** irrational

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the distance between $P(2, -1)$ and $Q(5, 3)$. 2

17. Find the zero of $3x - 18$ and show that it lies on the x-axis of the graph $y = 3x - 18$. 2

18. Express $0.\overline{3}$ as a fraction in lowest terms. 2

19. Expand $(3x + 5y)^2$. 2

20. Find the midpoint of the segment with endpoints $(-3, 7)$ and $(9, 1)$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Show that the points $A(0, 0)$, $B(4, 3)$ and $C(8, 0)$ are the vertices of an isosceles triangle. 3

22. Draw up a table for $y = 2x + 3$ for $x = -1, 0, 1, 2$ and state the slope and the y-intercept. 3

23. Express $0.\overline{36}$ in the form $\frac{p}{q}$. 3

24. Expand $(2x + 3y)^2$ and verify for $x = 1$, $y = 1$. 3

OR

Find the point on the x-axis which is equidistant from $A(-1, 4)$ and $B(5, -4)$. 3

25. The line $y = 2x + k$ has slope 2 and passes through $(3, 11)$. Find k and write the line. 3

OR

Express $0.1\overline{6}$ in the form $\frac{p}{q}$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Show that the points $A(0, 0)$, $B(4, 1)$, $C(6, 4)$ and $D(2, 3)$ taken in order form a parallelogram. 5

OR

The cost of x notebooks is a linear polynomial: a shop charges ₹60 for the bag plus ₹12 per notebook. Write the cost $C(x)$, find the cost of 5 notebooks and how many notebooks can be bought for ₹168. 5

27. Find the decimal expansion of $\frac{1}{7}$ by long division and state the length of the repeating block. 5

OR

Expand $(x + 2y + 3z)^2$. 5

END OF QUESTION PAPER Total: 50 Marks

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